BCS Theory

Hongquan

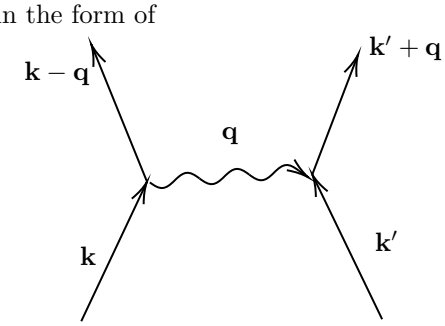
August 3, 2023

1 BCS Ground State

Consider the many-body Hamiltonian

$$H = \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} n_{\mathbf{k},\sigma} + \sum_{\mathbf{k},\mathbf{k}',\mathbf{q},\sigma,\sigma'} V_{\mathbf{k},\mathbf{k}',\mathbf{q}} c^\dagger_{\mathbf{k}'+\mathbf{q},\sigma'} c_{\mathbf{k}',\sigma'} c^\dagger_{\mathbf{k}-\mathbf{q},\sigma} c_{\mathbf{k},\sigma}. \tag{1}$$

The potential is essentially a scattering problem between two electrons mediated by a phonon with momentum $\mathbf{q}$. The Feynmann diagram is in the form of



The second quantization of small V gives

$$\langle \mathbf{k}-\mathbf{q}, \mathbf{k}'+\mathbf{q}|V|\mathbf{k},\mathbf{k}'\rangle = \frac{\hbar\omega_{\mathbf{q}}|M_{\mathbf{q}}|^2}{(\epsilon_{\mathbf{k}}-\epsilon_{\mathbf{k}-\mathbf{q}})^2-(\hbar\omega_{\mathbf{q}})^2} + \frac{4\pi e^2}{q^2+\lambda^2}, \tag{2}$$

with the last term being the screened Columb interaction. From this we see that this can only be attractive if $|\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}-\mathbf{q}}| < \hbar\omega_{\mathbf{q}}$.

We solve this problem in the following steps:

$$\boxed{\begin{array}{l} \text{1. Construct a many-body state of superpositions of Cooper pairs.} \\ \text{2. Investigate the properties of the Cooper pair creation operators.} \\ \text{3. Use variational principle to find the minimal energy of this state.} \end{array}} \tag{3}$$

Now consider a many-body state

$$|BCS\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow})\,|0\rangle\,. \tag{4}$$

For each $\mathbf{k}$, either the paired state is occupied or not, so $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$. For simplicity, we define

$$\begin{aligned} b^\dagger_{\mathbf{k}} &:= c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} \\ b_{\mathbf{k}} &:= c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \end{aligned} \tag{5}$$

are the creation and annihilation operators for a cooper pair. They satisfies the commutator relation:

$$\left[b_{\mathbf{k}}, b^\dagger_{\mathbf{k}'}\right] = c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} c^\dagger_{\mathbf{k}'\uparrow} c^\dagger_{-\mathbf{k}'\downarrow} - c^\dagger_{\mathbf{k}'\uparrow} c^\dagger_{-\mathbf{k}'\downarrow} c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow}. \tag{6}$$

The first term is

$$
\begin{aligned}
c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow}c^\dagger_{\mathbf{k}'\uparrow}c^\dagger_{-\mathbf{k}'\downarrow} &= c_{-\mathbf{k}\downarrow}(\delta_{\mathbf{k},\mathbf{k}'} - c^\dagger_{\mathbf{k}'\uparrow}c_{\mathbf{k}\uparrow})c^\dagger_{-\mathbf{k}'\downarrow} \\
&= \delta_{\mathbf{k},\mathbf{k}'}(1-n_{-\mathbf{k}\downarrow}) - c_{-\mathbf{k}\downarrow}c^\dagger_{\mathbf{k}'\uparrow}c_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}'\downarrow} \\
&= \delta_{\mathbf{k},\mathbf{k}'}(1-n_{-\mathbf{k}\downarrow}) + c_{-\mathbf{k}\downarrow}c^\dagger_{\mathbf{k}'\uparrow}c^\dagger_{-\mathbf{k}'\downarrow}c_{\mathbf{k}\uparrow} \\
&= \delta_{\mathbf{k},\mathbf{k}'}(1-n_{-\mathbf{k}\downarrow}) - c^\dagger_{\mathbf{k}'\uparrow}c_{-\mathbf{k}\downarrow}c^\dagger_{-\mathbf{k}'\downarrow}c_{\mathbf{k}\uparrow} \\
&= \delta_{\mathbf{k},\mathbf{k}'}(1-n_{-\mathbf{k}\downarrow}) - c^\dagger_{\mathbf{k}'\uparrow}(\delta_{\mathbf{k},\mathbf{k}'} - c^\dagger_{-\mathbf{k}'\downarrow}c_{-\mathbf{k}\downarrow})c_{\mathbf{k}\uparrow} \\
&= \delta_{\mathbf{k},\mathbf{k}'}(1-n_{-\mathbf{k}\downarrow}) - \delta_{\mathbf{k},\mathbf{k}'}n_{\mathbf{k}\uparrow} + c^\dagger_{\mathbf{k}'\uparrow}c^\dagger_{-\mathbf{k}'\downarrow}c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow}
\end{aligned} \tag{7}
$$

Hence

$$
\left[b_{\mathbf{k}}, b^\dagger_{\mathbf{k}'}\right] = \delta_{\mathbf{k},\mathbf{k}'}(1 - n_{\mathbf{k},\uparrow} - n_{-\mathbf{k},\downarrow}). \tag{8}
$$

Quite obviously

$$
\left[b^\dagger_{\mathbf{k}}, b^\dagger_{\mathbf{k}'}\right] = [b_{\mathbf{k}}, b_{\mathbf{k}'}] = 0. \tag{9}
$$

The number operator for b is

$$
\begin{aligned}
b^\dagger_{\mathbf{k}}b_{\mathbf{k}} = c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow} &= -c^\dagger_{-\mathbf{k}\downarrow}c^\dagger_{\mathbf{k}\uparrow}c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow} \\
&= c^\dagger_{-\mathbf{k}\downarrow}c_{-\mathbf{k}\downarrow}c^\dagger_{\mathbf{k}\uparrow}c_{\mathbf{k}\uparrow} = n_{-\mathbf{k}\downarrow}n_{\mathbf{k},\uparrow}.
\end{aligned} \tag{10}
$$

Now we may write the BCS states as

$$
|BCS\rangle = \prod_{\mathbf{k}}(u_{\mathbf{k}} + v_{\mathbf{k}}b^\dagger_{\mathbf{k}})\,|0\rangle\,. \tag{11}
$$

So we may calculate the total number of electrons by $\langle BCS|\sum_{\mathbf{k},\sigma}c^\dagger_{\mathbf{k},\sigma}c_{\mathbf{k},\sigma}\,|BCS\rangle$. However if $(\mathbf{k},\sigma)$ is occupied and since all electrons come in pairs, $(-\mathbf{k},-\sigma)$ is also occupied. This would imply that for BCS states,

$$
n_{\mathbf{k}\uparrow} + n_{\mathbf{k}\downarrow} = n_{\mathbf{k},\uparrow}n_{-\mathbf{k}\downarrow} + n_{\mathbf{k}\downarrow}n_{-\mathbf{k}\uparrow} = b^\dagger_{\mathbf{k}}b_{\mathbf{k}} + b^\dagger_{-\mathbf{k}}b_{-\mathbf{k}}. \tag{12}
$$

So

$$
\begin{aligned}
\langle N\rangle &= \langle BCS|\sum_{\mathbf{k},\sigma}c^\dagger_{\mathbf{k},\sigma}c_{\mathbf{k},\sigma}\,|BCS\rangle \\
&= \sum_{\mathbf{k}}\langle BCS|\,b^\dagger_{\mathbf{k}}b_{\mathbf{k}}\,|BCS\rangle + \sum_{-\mathbf{k}}\langle BCS|\,b^\dagger_{\mathbf{k}}b_{\mathbf{k}}\,|BCS\rangle \\
&= 2\sum_{\mathbf{k}}\langle 0|\prod_{\mathbf{k}'}(u^*_{\mathbf{k}'} + v^*_{\mathbf{k}'}b_{\mathbf{k}'})b^\dagger_{\mathbf{k}}b_{\mathbf{k}}\prod_{\mathbf{k}''}(u_{\mathbf{k}''} + v_{\mathbf{k}''}b^\dagger_{\mathbf{k}''})\,|0\rangle \\
&= 2\sum_{\mathbf{k}}\langle 0|\,(u^*_{\mathbf{k}} + v^*_{\mathbf{k}}b_{\mathbf{k}})b^\dagger_{\mathbf{k}}b_{\mathbf{k}}(u_{\mathbf{k}} + v_{\mathbf{k}}b^\dagger_{\mathbf{k}})\,|0\rangle \\
&= 2\sum_{\mathbf{k}}\langle 0|\,v^*_{\mathbf{k}}b_{\mathbf{k}}b^\dagger_{\mathbf{k}}b_{\mathbf{k}}v_{\mathbf{k}}b^\dagger_{\mathbf{k}}\,|0\rangle = 2\sum_{\mathbf{k}}|v_{\mathbf{k}}|^2.
\end{aligned} \tag{13}
$$

Up there we used the fact that

$$
(u^*_{\mathbf{k}} + v^*_{\mathbf{k}}b_{\mathbf{k}})(u_{\mathbf{k}} + v_{\mathbf{k}}b^\dagger_{\mathbf{k}})\,|0\rangle = (|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 b_{\mathbf{k}}b^\dagger_{\mathbf{k}})\,|0\rangle = |0\rangle\,. \tag{14}
$$

Now comes the important part: We show that for proper values of $v_{\mathbf{k}}$, this state could have less energy—being the ground state.

$$
\langle BCS|H|BCS\rangle = \langle BCS|\sum_{\mathbf{k},\sigma}\epsilon_{\mathbf{k}}n_{\mathbf{k},\sigma} + \sum_{\mathbf{k},\mathbf{k}',\mathbf{q},\sigma,\sigma'}V_{\mathbf{k},\mathbf{k}',\mathbf{q}}c^\dagger_{\mathbf{k}'+\mathbf{q},\sigma'}c_{\mathbf{k}',\sigma'}c^\dagger_{\mathbf{k}-\mathbf{q},\sigma}c_{\mathbf{k},\sigma}\,|BCS\rangle\,. \tag{15}
$$

Again for BCS states, when we destroy $(\mathbf{k},\sigma)$ we must also destroy $(-\mathbf{k},-\sigma)$ to still be a BCS state in order to have a non-zero inner product. So

$$
\mathbf{k}' = -\mathbf{k}; \quad \sigma' = -\sigma; \quad \mathbf{k}' + \mathbf{q} = -(\mathbf{k} - \mathbf{q}). \tag{16}
$$

Then the original sum becomes

$$\langle BCS| \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma} + \sum_{\mathbf{k},\mathbf{q},\sigma} V_{\mathbf{k},-\mathbf{k},\mathbf{q}} c^\dagger_{-\mathbf{k},-\sigma} c_{-\mathbf{k}+\mathbf{q},-\sigma} c^\dagger_{\mathbf{k}-\mathbf{q},\sigma} c_{\mathbf{k},\sigma} |BCS\rangle . \tag{17}$$

Let $\mathbf{k}-\mathbf{q} \equiv \mathbf{k}'$ and $V_{\mathbf{k},-\mathbf{k},\mathbf{q}} \equiv V_{\mathbf{k},\mathbf{k}'}$ then the reduced Hamiltonian becomes

$$\begin{aligned} H_{red} &= \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma} + \sum_{\mathbf{k},\mathbf{k}',\sigma} V_{\mathbf{k},\mathbf{k}'} c^\dagger_{-\mathbf{k},-\sigma} c_{-\mathbf{k}',-\sigma} c^\dagger_{\mathbf{k}',\sigma} c_{\mathbf{k},\sigma} \\ &= \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma} + 2 \sum_{\mathbf{k},\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} b^\dagger_{\mathbf{k}'} b_{\mathbf{k}}. \end{aligned} \tag{18}$$

Now let $\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \mu$, which is the distance to the Fermi sea, we have

$$\langle BCS| H - \mu |BCS\rangle = 2 \sum_{\mathbf{k}} \xi_{\mathbf{k}} |v_{\mathbf{k}}|^2 + 2 \sum_{\mathbf{k},\mathbf{k}'} \langle 0| \prod_\alpha (u^*_\alpha + v^*_\alpha b_\alpha) V_{\mathbf{k},\mathbf{k}'} b^\dagger_{\mathbf{k}'} b_{\mathbf{k}} \prod_\beta (u_\beta + v_\beta b^\dagger_\beta) |0\rangle . \tag{19}$$

The second term has non-zero summand only for

$$\begin{aligned} &\langle 0| (u^*_{\mathbf{k}} + v^*_{\mathbf{k}} b_{\mathbf{k}})(u^*_{\mathbf{k}'} + v^*_{\mathbf{k}'} b_{\mathbf{k}'}) V_{\mathbf{k},\mathbf{k}'} b^\dagger_{\mathbf{k}'} b_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} b^\dagger_{\mathbf{k}})(u_{\mathbf{k}'} + v_{\mathbf{k}'} b^\dagger_{\mathbf{k}'}) |0\rangle \\ &= V_{\mathbf{k},\mathbf{k}'} u^*_{\mathbf{k}} v^*_{\mathbf{k}'} v_{\mathbf{k}} u_{\mathbf{k}'}. \end{aligned} \tag{20}$$

So

$$\langle BCS| H - \mu |BCS\rangle = 2 \sum_{\mathbf{k}} \xi_{\mathbf{k}} |v_{\mathbf{k}}|^2 + 2 \sum_{\mathbf{k},\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} u^*_{\mathbf{k}} v^*_{\mathbf{k}'} v_{\mathbf{k}} u_{\mathbf{k}'}. \tag{21}$$

Now, for the ground state, we must minimize the free energy $F = W - TS$. At zero temperature, we assume $v_{\mathbf{k}}$ are real and we minimize this w.r.t. v_i then

$$\frac{\partial}{\partial v_i} \langle BCS| H - \mu |BCS\rangle = 4\xi_i v_i + 2 \sum_{\mathbf{k}'} V_{i,\mathbf{k}'} u_{\mathbf{k}'} v_{\mathbf{k}'} \frac{\partial}{\partial v_i} (\sqrt{1 - v_i^2} v_i) = 0. \tag{22}$$

Then

$$2\xi_i v_i + \sum_{\mathbf{k}'} V_{i,\mathbf{k}'} u_{\mathbf{k}'} v_{\mathbf{k}'} (u_i - \frac{v_i^2}{u_i}) = 0. \tag{23}$$

Define

$$\Delta_{\mathbf{k}} := -\sum_{\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} u_{\mathbf{k}'} v_{\mathbf{k}'} \tag{24}$$

then

$$2\xi_i = \Delta_i (\frac{1 - 2v_i^2}{v_i u_i}) = 0 \Rightarrow \frac{u_i v_i}{1 - 2v_i^2} = \frac{\Delta_i}{2\xi_i}. \tag{25}$$

Rearranging the above equation and take the square, then

$$(\Delta^2 + \xi^2) v^4 - (\Delta^2 + \xi^2) v^2 + \frac{\Delta^2}{4} = 0. \tag{26}$$

$$v^2 = \frac{1}{2} \left[1 \pm \frac{\xi}{\sqrt{\Delta^2 + \xi^2}} \right]. \tag{27}$$

Define $E_{\mathbf{k}} := \sqrt{\xi^2_{\mathbf{k}} + \Delta^2_{\mathbf{k}}}$ and that we know when $V = 0$ we must have $v = 0$—the free electron gas. So

$$v^2_{\mathbf{k}} = \frac{1}{2} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \right); \quad u^2_{\mathbf{k}} = \frac{1}{2} \left(1 + \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \Rightarrow u_{\mathbf{k}} v_{\mathbf{k}} = \Delta_{\mathbf{k}} / 2E_{\mathbf{k}}. \tag{28}$$

Now substituting the solution back to the definition of $\Delta_{\mathbf{k}}$ then we have

$$\Delta_{\mathbf{k}} = -\sum_{\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} \frac{\Delta_{\mathbf{k}'}}{2E_{\mathbf{k}'}} = -\sum_{\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} \frac{\Delta_{\mathbf{k}'}}{2\sqrt{\xi^2_{\mathbf{k}'} + \Delta^2_{\mathbf{k}'}}}. \tag{29}$$

And that our ground state energy is

$$\begin{aligned}\langle BCS|\, H-\mu\,|BCS\rangle &= 2\sum_{\mathbf{k}}\xi_{\mathbf{k}}|v_{\mathbf{k}}|^2 - 2\sum_{\mathbf{k}} u_{\mathbf{k}}v_{\mathbf{k}}\Delta_{\mathbf{k}} \\ &= 2\sum_{\mathbf{k}}\xi_{\mathbf{k}}\frac{1}{2}\left(1-\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right) - 2\sum_{\mathbf{k}}\frac{\Delta_{\mathbf{k}}}{2E_{\mathbf{k}}}\Delta_{\mathbf{k}} \\ &= \sum_{\mathbf{k}}\xi_{\mathbf{k}} - \frac{\xi_{\mathbf{k}}^2}{E_{\mathbf{k}}} - \frac{\Delta_{\mathbf{k}}^2}{E_{\mathbf{k}}} = \sum_{\mathbf{k}}\xi_{\mathbf{k}} - E_{\mathbf{k}}.\end{aligned} \tag{30}$$

It is extremely important to notice that, we are solving for the many-body state (like a configuration) of the system, instead of one state. So what reflect this state more is the density of states instead of some energy spectrum, since energy spectrum is for one electron. In the free electron gas case, the energy is $\sum_{\mathbf{k}}\xi_{\mathbf{k}}$.

Now we may look at excitations of the ground state $|BCS\rangle$. Recall that this ground state is

$$|BCS\rangle = \prod_{\mathbf{k}}(u_{\mathbf{k}} + v_{\mathbf{k}}b_{\mathbf{k}}^{\dagger})\,|0\rangle\,. \tag{31}$$

What we can do is to break one pair $b_{\mathbf{k}}$ and place the two unpaired electrons somewhere else in the $\mathbf{k}-$space. So the excited state seems to be

$$|Exc\rangle = c_{-\mathbf{k}_2\downarrow}^{\dagger}c_{\mathbf{k}_1\uparrow}^{\dagger}\prod_{\mathbf{k}\neq\mathbf{k}_1,\mathbf{k}_2}(u_{\mathbf{k}} + v_{\mathbf{k}}b_{\mathbf{k}}^{\dagger})\,|0\rangle\,. \tag{32}$$

First of all, we must check that this excited state is orthogonal to the ground state:

$$\begin{aligned}\langle BCS|Exc\rangle &= \langle 0|\prod_{\mathbf{k}}(u_{\mathbf{k}} + v_{\mathbf{k}}b_{\mathbf{k}})c_{-\mathbf{k}_2\downarrow}^{\dagger}c_{\mathbf{k}_1\uparrow}^{\dagger}\prod_{\mathbf{k}\neq\mathbf{k}_1,\mathbf{k}_2}(u_{\mathbf{k}} + v_{\mathbf{k}}b_{\mathbf{k}}^{\dagger})\,|0\rangle \\ &= \langle 0|\,(u_{\mathbf{k}_2} + v_{\mathbf{k}_2}b_{\mathbf{k}_2})(u_{\mathbf{k}_1} + v_{\mathbf{k}_1}b_{\mathbf{k}_1})c_{-\mathbf{k}_2\downarrow}^{\dagger}c_{\mathbf{k}_1\uparrow}^{\dagger}\,|0\rangle \\ &= u_{\mathbf{k}_2}u_{\mathbf{k}_1}\,\langle 0|\,c_{-\mathbf{k}_2\downarrow}^{\dagger}c_{\mathbf{k}_1\uparrow}^{\dagger}\,|0\rangle = 0.\end{aligned} \tag{33}$$

Now comes the one key of the BCS theory: this is exactly an excited state—of a higher total energy. It takes a long computation to compute $\langle Exc|H-\mu|Exc\rangle$ again, which is similar to what we did above. Here we provide with a more "physical" approach: First we look at the expression of $\langle BCS|H-\mu|BCS\rangle$, which is just

$$2\sum_{\mathbf{k}}\xi_{\mathbf{k}}v_{\mathbf{k}}^2 + 2\sum_{\mathbf{k},\mathbf{k}'}V_{\mathbf{k},\mathbf{k}'}u_{\mathbf{k}}v_{\mathbf{k}'}v_{\mathbf{k}}u_{\mathbf{k}'}. \tag{34}$$

Now for the excited state, the two escaping electrons cannot interact with the other Cooper pairs, and hence they have no potential energy. So the energy of that excited state is

$$2\sum_{\mathbf{k}\neq\mathbf{k}_1,\mathbf{k}_2}\xi_{\mathbf{k}}v_{\mathbf{k}}^2 + 2\sum_{\mathbf{k},\mathbf{k}'\neq\mathbf{k}_1,\mathbf{k}_2}V_{\mathbf{k},\mathbf{k}'}u_{\mathbf{k}}v_{\mathbf{k}'}v_{\mathbf{k}}u_{\mathbf{k}'} + \xi_{\mathbf{k}_1} + \xi_{\mathbf{k}_2}. \tag{35}$$

So the energy difference $\langle Exc|H-\mu|Exc\rangle - \langle BCS|H-\mu|BCS\rangle$ is

$$\xi_{\mathbf{k}_1}(1-2v_{\mathbf{k}_1}^2) + \xi_{\mathbf{k}_2}(1-2v_{\mathbf{k}_1}^2) - 2\sum_{\mathbf{k}}V_{\mathbf{k},\mathbf{k}_1}u_{\mathbf{k}}v_{\mathbf{k}}u_{\mathbf{k}_1}v_{\mathbf{k}_1} - 2\sum_{\mathbf{k}}V_{\mathbf{k},\mathbf{k}_2}u_{\mathbf{k}}v_{\mathbf{k}}u_{\mathbf{k}_2}v_{\mathbf{k}_2}. \tag{36}$$

Using the above expressions for Δ and E, we have the above difference equals

$$\frac{\xi_{\mathbf{k}_1}}{E_{\mathbf{k}_1}}\xi_{\mathbf{k}_1} + \frac{\xi_{\mathbf{k}_2}}{E_{\mathbf{k}_2}}\xi_{\mathbf{k}_2} + 2\Delta_{\mathbf{k}_1}\frac{\Delta_{\mathbf{k}_1}}{2E_{\mathbf{k}_1}} + 2\Delta_{\mathbf{k}_2}\frac{\Delta_{\mathbf{k}_2}}{2E_{\mathbf{k}_2}} = E_{\mathbf{k}_1} + E_{\mathbf{k}_2}. \tag{37}$$

Therefore the excitation energy is $E_{\mathbf{k}}$ for an electron at $\mathbf{k}$ in a broken pair. Since a smallest excitation consists of two electrons in a broken pair, the minimum excitation energy should be $2\Delta_{\min} \leq \Delta_{\mathbf{k}_1} + \Delta_{\mathbf{k}_2}$.

Now, suppose the energy is non-zero and we minimize $W - TS$. To do this simpler, we may harness the following section.

2 Temperature Dependence

We may calculate:

$$(uc^\dagger_{\mathbf{k}\uparrow} - vc_{-\mathbf{k}\downarrow})(u + vb^\dagger_{\mathbf{k}}) = u^2 c^\dagger_{\mathbf{k}\uparrow} + uvc^\dagger_{\mathbf{k}\uparrow} b^\dagger_{\mathbf{k}} - uvc_{-\mathbf{k}\downarrow} - v^2 c_{-\mathbf{k}\downarrow} b^\dagger_{\mathbf{k}}, \tag{38}$$

and that

$$c^\dagger_{\mathbf{k}\uparrow} b^\dagger_{\mathbf{k}} = 0; \quad c_{-\mathbf{k}\downarrow} |0\rangle = 0; \quad c_{-\mathbf{k}\downarrow} b^\dagger_{\mathbf{k}} |0\rangle = c_{-\mathbf{k}\downarrow} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} |0\rangle = -c^\dagger_{\mathbf{k}\uparrow} |0\rangle . \tag{39}$$

So we find that

$$(uc^\dagger_{\mathbf{k}\uparrow} - vc_{-\mathbf{k}\downarrow})(u + vb^\dagger_{\mathbf{k}}) |0\rangle = (u^2 + v^2) c^\dagger_{\mathbf{k}\uparrow} |0\rangle = c^\dagger_{\mathbf{k}\uparrow} |0\rangle . \tag{40}$$

On defining $\gamma^\dagger_{\mathbf{k}\uparrow} := u_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} - v_{\mathbf{k}} c_{-\mathbf{k}\downarrow}$, and a similar operator $\gamma^\dagger_{\mathbf{k}\downarrow} := u_{\mathbf{k}} c^\dagger_{\mathbf{k}\downarrow} + v_{\mathbf{k}} c_{-\mathbf{k}\uparrow}$ we see that the excited state is in fact,

$$|Exc\rangle = \gamma^\dagger_{\mathbf{k}_2\downarrow} \gamma^\dagger_{\mathbf{k}_1\uparrow} |BCS\rangle . \tag{41}$$

We may view $\gamma^\dagger_{\mathbf{k},\sigma}$ as a creation operator creating a quasi-particle and the energy spectrum of this is $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}} + \Delta_{\mathbf{k}}}$. Such quasi-particle has an energy gap of $\Delta_{\mathbf{k}}$.

Therefore this reduced Hamiltonian can be written as

$$H = \sum_{\mathbf{k},\sigma} E_{\mathbf{k}} \gamma^\dagger_{\mathbf{k},\sigma} \gamma_{\mathbf{k},\sigma} + W_0, \tag{42}$$

where W_0 is the energy of the BCS state.

This quasi-particle is called the Bogoliubon, we can easily check its commutator relations. First we look at their expressions

$$\begin{aligned} \gamma^\dagger_{\mathbf{k}\uparrow} &= u_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} - v_{\mathbf{k}} c_{-\mathbf{k}\downarrow} \\ \gamma^\dagger_{\mathbf{k}\downarrow} &= u_{\mathbf{k}} c^\dagger_{\mathbf{k}\downarrow} + v_{\mathbf{k}} c_{-\mathbf{k}\uparrow} \\ \gamma_{\mathbf{k}\uparrow} &= u_{\mathbf{k}} c_{\mathbf{k}\uparrow} - v_{\mathbf{k}} c^\dagger_{-\mathbf{k}\downarrow} \\ \gamma_{\mathbf{k}\downarrow} &= u_{\mathbf{k}} c_{\mathbf{k}\downarrow} + v_{\mathbf{k}} c^\dagger_{-\mathbf{k}\uparrow}. \end{aligned} \tag{43}$$

It is easily checked that such operators are fermionic. Now, at non-zero temperature, we must consider the thermodynamic probability that such excitations are made. Since such quasi-particles are fermions, they obey the Fermi-Dirac distribution:

$$f_{\mathbf{k}} = \frac{1}{1 + e^{\beta E_{\mathbf{k}}}}. \tag{44}$$

Now we may compute the Free energy. First we look at the energy term W: There is a $f_{\mathbf{k}}$ chance to create a quasi-particle and the emergence of one such quasi-particle leads to the emergence of another quasiparicle. While each excitation is produced by two such quasi-particles, it can be triggered by the emergence of any one of them. Hence the probability of an excitation is $2f_{\mathbf{k}}$, and each excitation contributes to $\xi_{\mathbf{k}}$ *kinetic* energy. So the kinetic energy is

$$W_K = 2 \sum_{\mathbf{k}} \xi_{\mathbf{k}} v^2_{\mathbf{k}} (1 - 2f_{\mathbf{k}}) + 2f_{\mathbf{k}} \xi_{\mathbf{k}}. \tag{45}$$

Now similarly the interaction energy is

$$W_I = 2 \sum_{\mathbf{k},\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} u_{\mathbf{k}} v_{\mathbf{k}'} v_{\mathbf{k}} u_{\mathbf{k}'} (1 - 2f_{\mathbf{k}})(1 - 2f_{\mathbf{k}'}). \tag{46}$$

The entropy is given by $k_B \ln \Omega$ while Ω is the number of possible many-body states. This is equivalent to the number of possible excitations of the ground state:

$$-TS = k_B T [f_{\mathbf{k}} \ln f_{\mathbf{k}} + (1 - f_{\mathbf{k}}) \ln(1 - f_{\mathbf{k}})]. \tag{47}$$

Quite similarly, we minimize this free energy:

$$0 = \frac{\partial}{\partial v_i}(W_K + W_I - TS) = 2\xi_i v_i (1 - 2f_{\mathbf{k}}) + (1 - 2f_{\mathbf{k}}) \sum_{\mathbf{k}'} V_{i,\mathbf{k}'} u_{\mathbf{k}'} v_{\mathbf{k}'} (u_i - \frac{v_i^2}{u_i})(1 - 2f_{\mathbf{k}'}). \tag{48}$$

So

$$2\xi_i v_i + \sum_{\mathbf{k}'} V_{i,\mathbf{k}'} u_{\mathbf{k}'} v_{\mathbf{k}'} (u_i - \frac{v_i^2}{u_i})(1 - 2f_{\mathbf{k}'}) = 0. \tag{49}$$

We define the gap function $\Delta_{\mathbf{k}}$ to be

$$\Delta_{\mathbf{k}} := -\sum_{\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} u_{\mathbf{k}'} v_{\mathbf{k}'} (1 - 2f_{\mathbf{k}'}) \tag{50}$$

then everything else are identical to the zero-temperature case. And now our self-consistent equation becomes

$$\Delta_{\mathbf{k}} := -\sum_{\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} \frac{\Delta_{\mathbf{k}'}}{2E_{\mathbf{k}'}} (1 - \frac{2}{1 + e^{\beta E_{\mathbf{k}'}}}) = -\sum_{\mathbf{k}'} V_{\mathbf{k},\mathbf{k}'} \frac{\Delta_{\mathbf{k}'}}{2E_{\mathbf{k}'}} \tanh\left(\frac{E_{\mathbf{k}'}}{2k_B T}\right). \tag{51}$$

3 Multiorbital

Now the main difference is that the interaction energy is different:

$$H = \sum_{\mathbf{k},l\sigma} \epsilon_{\mathbf{k},l} n_{\mathbf{k},l\sigma} + \sum_{l_1,l_2,l_3,l_4} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q},\sigma,\sigma'} [V_{\mathbf{k},\mathbf{k}',\mathbf{q}}]_{l_1 l_2, l_3 l_4} c^\dagger_{\mathbf{k}'+\mathbf{q},l_1\sigma'} c_{\mathbf{k}',l_2\sigma'} c^\dagger_{\mathbf{k}-\mathbf{q},l_3\sigma} c_{\mathbf{k},l_4\sigma}. \tag{52}$$

Of course, now we must define

$$b^\dagger_{\mathbf{k},l_1 l_2} := c^\dagger_{\mathbf{k}\uparrow,l_1} c^\dagger_{-\mathbf{k}\downarrow,l_2}. \tag{53}$$

Along with it we have $u_{l_1 l_2}$ and $v_{l_1 l_2}$.

Similarly, we reduce this Hamiltonian to

$$H_{red} = \sum_{\mathbf{k},\sigma,l} \epsilon_{\mathbf{k},l} c^\dagger_{\mathbf{k},\sigma,l} c_{\mathbf{k},\sigma,l} + 2 \sum_{\mathbf{k},\mathbf{k}',\{l\}} (V_{\mathbf{k},\mathbf{k}'})_{l_1 l_2, l_3 l_4} b^\dagger_{\mathbf{k}',l_1 l_2} b_{\mathbf{k},l_3 l_4}. \tag{54}$$

And further to

$$W_0 = \sum_{\mathbf{k},\sigma,l} 2\epsilon_{\mathbf{k},l} |v_{\mathbf{k},ll}|^2 + 2 \sum_{\mathbf{k},\mathbf{k}',\{l\}} (V_{\mathbf{k},\mathbf{k}'})_{l_1 l_2, l_3 l_4} u_{\mathbf{k}',l_1 l_2} v_{\mathbf{k}',l_1 l_2} u_{\mathbf{k},l_3 l_4} v_{\mathbf{k}',l_3 l_4}. \tag{55}$$

Now we minimize this energy:

$$\partial_{v_{\mathbf{k},ij}} W_0 = 4\epsilon_{\mathbf{k},i} v_{\mathbf{k},ij} \delta_{ij} + 2 \sum_{\mathbf{k}',l_1,l_2} (V_{\mathbf{k},\mathbf{k}'})_{l_1 l_2, ij} u_{\mathbf{k}',l_1 l_2} v_{\mathbf{k}',l_1 l_2} (u_{\mathbf{k},ij} - \frac{v^2_{\mathbf{k},ij}}{u_{\mathbf{k},ij}}). \tag{56}$$

Again we may define

$$\Delta_{\mathbf{k},ij} := -\sum_{\mathbf{k}',l_1,l_2} (V_{\mathbf{k},\mathbf{k}'})_{l_1 l_2, ij} u_{\mathbf{k}',l_1 l_2} v_{\mathbf{k}',l_1 l_2} \tag{57}$$

then

$$4\epsilon_{\mathbf{k},i} v_{\mathbf{k},ij} \delta_{ij} - 2\Delta_{\mathbf{k},ij} (u_{\mathbf{k},ij} - \frac{v^2_{\mathbf{k},ij}}{u_{\mathbf{k},ij}}) = 0 \tag{58}$$

and for $i = j$, we have

$$\Delta_{\mathbf{k},i} = -\sum_{\mathbf{k}',j} (V_{\mathbf{k},\mathbf{k}'})_{jj,ii} \frac{\Delta_{\mathbf{k}',j}}{2E_{\mathbf{k}',j}}. \tag{59}$$

One may define $\Gamma_{i,j} := V_{jj,ii}$ and adding the temperature dependence we have

$$\Delta_{\mathbf{k},i} = -\sum_{\mathbf{k}',j} \Gamma_{i,j} \frac{\Delta_{\mathbf{k}',j}}{2E_{\mathbf{k}',j}} \tanh\left(\frac{E_{\mathbf{k}',j}}{2k_B T}\right). \tag{60}$$

4 Linearized Eliashberg Equation

4.1 Calculation of χ_0, χ_s and χ_c

The expression of χ_0 is given by

$$\chi_0(q) = -\sum_k G^{(0)}(k+q)G^{(0)}(k), \tag{61}$$

where

$$k = (\mathbf{k}, i\omega_n); \quad q = (\mathbf{q}, i\Omega_m), \quad \sum_k = \frac{T}{N_{\mathbf{k}}}\sum_n \sum_{\mathbf{k}}. \tag{62}$$

Then we define χ_s and χ_c to be the solution to :

$$\chi_s = \chi_0 + \chi_s U \chi_0; \tag{63}$$

$$\chi_c = \chi_0 - \chi_c U \chi_0. \tag{64}$$

4.1.1 Single Orbital

In the single-band case,

$$G^{(0)}(k) = \frac{1}{i\omega_n - \epsilon(\mathbf{k})}. \tag{65}$$

Let us first calculate

$$\sum_n \frac{1}{i\omega_n - \epsilon(\mathbf{k})} \frac{1}{i\Omega_m + i\Omega_m - \epsilon(\mathbf{k}+\mathbf{q})}. \tag{66}$$

To simplify, we consider the following integral

$$\int_\gamma n_F(z) \frac{1}{z-a}\frac{1}{z-b} dz, \quad \gamma = \lim_{R\to\infty}\{|z| = R\}, \tag{67}$$

where

$$n_F(z) = \frac{1}{\exp(z/T)+1}. \tag{68}$$

The auxiliary function n gives poles at $z = (2n+1)\pi i T = i\omega_n$, allowing us to harness the residue theorem. By residue theorem, we have

$$\frac{1}{2\pi i}\int_\gamma n_F(z)\frac{1}{z-a}\frac{1}{z-b}dz = \sum_n \mathrm{Res}(n, i\omega_n)\frac{1}{i\omega_n - a}\frac{1}{i\omega_n - b} + \frac{n_F(a)}{a-b} + \frac{n_F(b)}{b-a}. \tag{69}$$

And the residues are

$$\mathrm{Res}(n, i\omega_n) = \lim_{z\to i\omega_n}\frac{z - i\omega_n}{\exp(z/T)+1} = \frac{1}{\exp(i\omega_n/T)/T} = -T. \tag{70}$$

Therefore

$$\frac{1}{2\pi i}\int_\gamma n_F(z)\frac{1}{z-a}\frac{1}{z-b}dz = -T\sum_n \frac{1}{i\omega_n - a}\frac{1}{i\omega_n - b} + \frac{n_F(b) - n_F(a)}{b-a}. \tag{71}$$

Now, for $|z|$ large enough,

$$\left|\int_\gamma n_F(z)\frac{1}{z-a}\frac{1}{z-b}dz\right| \le \frac{1}{4}\left|\int_\gamma n_F(z)dz\right|\frac{1}{|z|^2}, \tag{72}$$

while

$$\int_\gamma n_F(z)dz = \sum_n 2\pi i \mathrm{Res}(n, i\omega_n) = -2\pi i T \#\{n : |\omega_n| \le |z|\}, \tag{73}$$

and

$$\#\{n : |\omega_n| \leq |z|\} = 2\#\{n : 0 < (2n+1)\pi T \leq |z|\} = \frac{|z|}{\pi T} + C,\ C \text{ is some constant.} \tag{74}$$

Then

$$\int_\gamma n_F(z)dz = -2i|z| + C. \tag{75}$$

Therefore

$$\left|\int_\gamma n_F(z)\frac{1}{z-a}\frac{1}{z-b}dz\right| \leq \frac{1}{2|z|} \to 0. \tag{76}$$

Hence

$$\sum_n \frac{1}{i\omega_n - a}\frac{1}{i\omega_n - b} = \frac{1}{T}\frac{n_F(b) - n_F(a)}{b-a}. \tag{77}$$

Our expression for χ_0 now is

$$\chi_0(q) = -\frac{1}{N_\mathbf{k}}\sum_\mathbf{k}\frac{n_F(\epsilon_{\mathbf{k}+\mathbf{q}} - i\Omega_m) - n_F(\epsilon_\mathbf{k})}{\epsilon_{\mathbf{k}+\mathbf{q}} - i\Omega_m - \epsilon_\mathbf{k}}. \tag{78}$$

Notice that $\Omega_m = 2\pi m T$, we have

$$n_F(\epsilon - i\Omega_m) = \frac{1}{\exp(\epsilon - i\Omega_m)/T + 1} = \frac{1}{\exp(\epsilon/T)\exp(-2\pi m i T/T) + 1} = \frac{1}{\exp(\epsilon/T) + 1} = n_F(\epsilon). \tag{79}$$

Then the above expression for χ_0 can be reduced to

$$\chi_0(q) = -\frac{1}{N_\mathbf{k}}\sum_\mathbf{k}\frac{n_F(\epsilon_{\mathbf{k}+\mathbf{q}}) - n_F(\epsilon_\mathbf{k})}{\epsilon_{\mathbf{k}+\mathbf{q}} - i\Omega_m - \epsilon_\mathbf{k}}. \tag{80}$$

Similarly, from the expression of ϕ_0, we can deduce that

$$\phi_0(q) = \sum_k G^{(0)}(q-k)G^{(0)}(k) = \sum_k \frac{1}{i\omega_n - \epsilon(\mathbf{k})}\frac{1}{-i\Omega_m + i\Omega_m - \epsilon(-\mathbf{k}+\mathbf{q})} \tag{81}$$

$$= -\sum_k \frac{1}{i\omega_n - \epsilon(\mathbf{k})}\frac{1}{i\Omega_m - i\Omega_m + \epsilon(\mathbf{q}-\mathbf{k})} = -\frac{1}{N_\mathbf{k}}\sum_\mathbf{k}\frac{n_F(-\epsilon_{\mathbf{q}-\mathbf{k}}) - n_F(\epsilon_\mathbf{k})}{i\Omega_m - \epsilon_{\mathbf{q}-\mathbf{k}} - \epsilon_\mathbf{k}}. \tag{82}$$

Quite directly, we have

$$\chi_s(1 - U\chi_0) = \chi_0 \Rightarrow \chi_s = \frac{\chi_0}{1 - U\chi_0}. \tag{83}$$

And the Stoner criterion is obtained when $\boxed{\chi_s \to \infty \Leftrightarrow U\chi_0 = 1}$.

Of course similarly

$$\chi_c = \frac{\chi_0}{1 + U\chi_0}. \tag{84}$$

There are several properties of χ_0: When β is very small (very high temperature),

$$n(\epsilon) = \frac{1}{e^{\beta\epsilon} + 1} \simeq \frac{1}{2 + \beta\epsilon} \simeq \frac{1}{2} - \frac{\beta}{4}\epsilon + O(\epsilon^2). \tag{85}$$

So the occupation function is almost linear. This would mean

$$-\frac{n_{\mathbf{k}+\mathbf{q}} - n_\mathbf{k}}{\epsilon_{\mathbf{k}+\mathbf{q}} - \epsilon_\mathbf{k}} = \frac{\beta}{4}, \tag{86}$$

a constant. So χ_0 as its average should also be that value and is close to 0.

When the $\mathbf{k}$ states are all occupied or unoccupied, the numerator is always 0 so χ_0 is zero.

4.1.2 Multi Orbital

In the multiorbital case,

$$G^0(k) = [i\omega_n I - H(\mathbf{k})]^{-1}. \tag{87}$$

Suppose $H(\mathbf{k})$ is diagonalized by a $U(n)$ matrix $a(\mathbf{k})$ with eigenvalues $\epsilon_j(\mathbf{k})$. Then $i\omega_n I - H(\mathbf{k})$ can also be diagonalized by $a(\mathbf{k})$ with eigenvalues $i\omega_n - \epsilon_j(\mathbf{k})$. Furthermore, it is direct to see that $G^0(k)$ is still diagonalized by $a(\mathbf{k})$ with eigenvalues $[i\omega_n - \epsilon_j(\mathbf{k})]^{-1}$ because

$$(aEa^\dagger)(aE^{-1}a^\dagger) = I. \tag{88}$$

So we may first generalize χ_0 to be

$$\chi_0(q)^{i_2 i_1}_{i_4 i_3} = -\sum_k [G^{(0)}(k+q)]^{i_1}{}_{i_3} [G^{(0)}(k)]_{i_4}{}^{i_2} \tag{89}$$

$$= -\sum_k \{a(\mathbf{k}+\mathbf{q})[i(\omega_n+\Omega_m)I - E(\mathbf{k}+\mathbf{q})]^{-1} a^\dagger(\mathbf{k}+\mathbf{q})\}^{i_1}{}_{i_3} \{a(\mathbf{k})[i\omega_n I - E(\mathbf{k})]^{-1} a^\dagger(\mathbf{k})\}_{i_4}{}^{i_2} \tag{90}$$

$$= -\sum_{k,j_1,j_2} [a_{\mathbf{k}+\mathbf{q}}]^{i_1}{}_{j_1} [i(\omega_n+\Omega_m)I - E_{\mathbf{k}+\mathbf{q}}]^{-1}_{j_1} [a^\dagger_{\mathbf{k}+\mathbf{q}}]^{j_1}{}_{i_3} [a_{\mathbf{k}}]_{i_4}{}^{j_2} [i(\omega_n+\Omega_m)I - E_{\mathbf{k}}]^{-1}_{j_2} [a^\dagger_{\mathbf{k}}]_{j_2}{}^{i_2} \tag{91}$$

$$= -\frac{T}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{j_1,j_2} [a_{\mathbf{k}+\mathbf{q}}]^{i_1}{}_{j_1} [a_{\mathbf{k}+\mathbf{q}}]^{*}_{i_3}{}^{j_1} [a_{\mathbf{k}}]_{i_4}{}^{j_2} [a_{\mathbf{k}}]^{*i_2}{}_{j_2} \sum_n \frac{1}{i(\omega_n+\Omega_m) - (\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1}} \frac{1}{i\omega_n - (\epsilon_{\mathbf{k}})_{j_2}} \tag{92}$$

Now using our previous identity we have

$$\chi_0(q)^{i_2 i_1}_{i_4 i_3} = -\frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{j_1,j_2} [a_{\mathbf{k}+\mathbf{q}}]^{i_1}{}_{j_1} [a_{\mathbf{k}+\mathbf{q}}]^{*}_{i_3}{}^{j_1} [a_{\mathbf{k}}]_{i_4}{}^{j_2} [a_{\mathbf{k}}]^{*i_2}{}_{j_2} \frac{n_F[(\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1} - i\Omega_m] - n_F[(\epsilon_{\mathbf{k}})_{j_2}]}{(\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1} - i\Omega_m - (\epsilon_{\mathbf{k}})_{j_2}}. \tag{93}$$

When $\Omega_m = 0$, the identity becomes

$$\chi_0(q)^{i_2 i_1}_{i_4 i_3} = -\frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{j_1,j_2} [a_{\mathbf{k}+\mathbf{q}}]^{i_1}{}_{j_1} [a_{\mathbf{k}+\mathbf{q}}]^{*}_{i_3}{}^{j_1} [a_{\mathbf{k}}]_{i_4}{}^{j_2} [a_{\mathbf{k}}]^{*i_2}{}_{j_2} \frac{n_F[(\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1}] - n_F[(\epsilon_{\mathbf{k}})_{j_2}]}{(\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1} - (\epsilon_{\mathbf{k}})_{j_2}}. \tag{94}$$

And the multiplication between the four-index matrices becomes

$$(AB)^{ij}_{kl} = \sum_{pq} A^{ij}_{pq} B^{pq}_{kl}. \tag{95}$$

Then we have

$$\chi_s = \chi_0 + \chi_s U \chi_0 \Rightarrow (1 - U\chi_0)\chi_s = \chi_0. \tag{96}$$

For a well-defined χ_s, $(I - U\chi_0)$ must be invertible. That is to say, $\det(I - U\chi_0) \neq 0$. Since $\det(I - U\chi_0)$ is a continuous function of U and that $\det(I - U\chi_0)\Big|_{U=0} = 1 > 0$, so if $\det(I - U\chi_0) < 0$ this would indicate that $U > U_c$. In other words, if $(I - U\chi_0)$ is not invertible, it will have an eigenvector v corresponding to eigenvalue 0. Then if $(1 - U\chi_0)\chi_s = \chi_0$ holds for some χ_s, it will also hold for $\chi_s + \alpha[v, v, ..., v]$ with arbitrary α, even making χ_s divergent.

4.2 Single Orbital

First we consider only singlets and single orbital models. The BCS theory provides us with the self-consistent equation of the order parameter $\Delta_{\mathbf{k}}$:

$$\Delta_{\mathbf{k}} = -\frac{1}{N} \sum_{\mathbf{k}'} V^{sing}_{\mathbf{k},\mathbf{k}'} \frac{\Delta_{\mathbf{k}'}}{2E_{\mathbf{k}'}} \tanh\left(\frac{\beta E_{\mathbf{k}'}}{2}\right) \tag{97}$$

where $E=\sqrt{\xi^2+\Delta^2}$. In the RPA context, the interaction is given by

$$\begin{aligned} V_{\mathbf{k},\mathbf{k}'}^{sing} &= \frac{1}{2}\left[V\left(\mathbf{k},\mathbf{k}'\right)+V\left(-\mathbf{k},\mathbf{k}'\right)\right] \\ &= \frac{1}{2}\left[2U+\frac{3}{2}U^2\chi_s\left(\mathbf{k}-\mathbf{k}'\right)-\frac{1}{2}U^2\chi_c\left(\mathbf{k}-\mathbf{k}'\right)+\frac{3}{2}U^2\chi_s\left(\mathbf{k}+\mathbf{k}'\right)-\frac{1}{2}U^2\chi_c\left(\mathbf{k}+\mathbf{k}'\right)\right] \end{aligned} \tag{98}$$

Here

$$V\left(\mathbf{k},\mathbf{k}'\right):=U+\frac{3}{2}U^2\chi_s\left(\mathbf{k}-\mathbf{k}'\right)-\frac{1}{2}U^2\chi_c\left(\mathbf{k}-\mathbf{k}'\right). \tag{99}$$

In the single-band case,

$$\chi_0(i\Omega_m,\mathbf{q})=-\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}}\frac{n_F(\epsilon_{\mathbf{k}+\mathbf{q}})-n_F(\epsilon_{\mathbf{k}})}{\epsilon_{\mathbf{k}+\mathbf{q}}-i\Omega_m-\epsilon_{\mathbf{k}}},\quad \chi_0(\mathbf{q})=\chi_0(0,\mathbf{q}). \tag{100}$$

with

$$\chi_s=\frac{\chi_0}{1-U\chi_0},\quad \chi_c=\frac{\chi_0}{1+U\chi_0}. \tag{101}$$

Since there is $\Delta_{\mathbf{k}}$ dependence in the denominator in Eq.(1), it is difficult to solve. The paper believed that around T_c, $\Delta_{\mathbf{k}}$ is small enough to adopt the approximation that $E_{\mathbf{k}}\simeq\xi_{\mathbf{k}}$. Then the original equation is reduced to

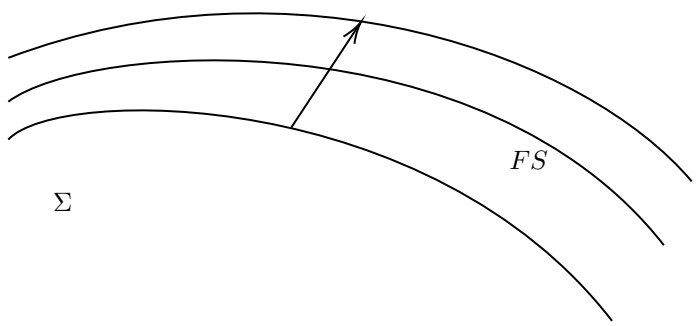
$$\Delta_{\mathbf{k}}=-\frac{1}{2N}\sum_{\mathbf{k}'}\frac{V_{\mathbf{k},\mathbf{k}'}^{sing}}{|\xi_{\mathbf{k}'}|}\Delta_{\mathbf{k}'}\tanh\left(\frac{|\xi_{\mathbf{k}'}|}{2k_BT_c}\right). \tag{102}$$

Of course we may approximate this by transforming the sum into an integral:

$$\Delta_{\mathbf{k}}=-\frac{1}{2N}\left(\frac{N}{(2\pi)^d}\int d\mathbf{k}'\right)\frac{V_{\mathbf{k},\mathbf{k}'}^{sing}}{|\xi_{\mathbf{k}'}|}\Delta_{\mathbf{k}'}\tanh\left(\frac{|\xi_{\mathbf{k}'}|}{2k_BT_c}\right). \tag{103}$$

The paper imposed that $V^{sing}\neq 0$ only when $|\xi_{\mathbf{k}'}|\leq\xi_c$. Therefore instead of integrating over BZ, we integrate over a tubular neighbourhood Σ of the FS:

$$\Sigma=\{\mathbf{k}\in BZ:|\xi_{\mathbf{k}}|\leq\xi_c\}. \tag{104}$$



For each $\mathbf{k}$ on the FS, we integrate along its perpendicular direction in Σ. If ξ_c is small enough, the integrand along that direction can be approximated by a constant if it is not rapidly changing, and the length of that arrow can be approximated by

$$\frac{2\xi_c}{|\Delta E/\Delta\mathbf{k}|}\approx\frac{2\xi_c}{|v_{\mathbf{k}}|}. \tag{105}$$

Since $1/|\xi_{\mathbf{k}}|$ is rapidly changing, we do not take it as constant so we factor that out, and the integral becomes

$$-\frac{1}{(2\pi)^d}\int_\Sigma d\mathbf{k}'\frac{V_{\mathbf{k},\mathbf{k}'}^{sing}}{|\xi_{\mathbf{k}'}|}\Delta_{\mathbf{k}'}\tanh\left(\frac{|\xi_{\mathbf{k}'}|}{2k_BT_c}\right)=-\frac{1}{(2\pi)^d}\int_\Sigma d\mathbf{k}'V_{\mathbf{k},\mathbf{k}'}^{sing}\Delta_{\mathbf{k}'}\int dx\delta(x-\xi_{\mathbf{k}'})\frac{\tanh\left(|\xi_{\mathbf{k}'}|/2k_BT_c\right)}{|\xi_{\mathbf{k}'}|} \tag{106}$$

This further equals

$$
\begin{aligned}
&-\frac{\xi_c}{(2\pi)^d}\int_{FS} d\mathbf{k}'\frac{V^{sing}_{\mathbf{k},\mathbf{k}'}}{|v_{\mathbf{k}'}|}\Delta_{\mathbf{k}'}\int dx\delta(x-\xi_{\mathbf{k}'})\frac{\tanh\left(|x|/2k_BT_c\right)}{|x|}\\
\approx&-\frac{\xi_c}{(2\pi)^d}\int_{FS} d\mathbf{k}'\frac{V^{sing}_{\mathbf{k},\mathbf{k}'}}{|v_{\mathbf{k}'}|}\Delta_{\mathbf{k}'}\int_{-\xi_c}^{\xi_c} dx\frac{1}{2\xi_c}\frac{\tanh\left(|x|/2k_BT_c\right)}{|x|}\\
=&-\frac{1}{(2\pi)^d}\int_{FS} d\mathbf{k}'\frac{V^{sing}_{\mathbf{k},\mathbf{k}'}}{|v_{\mathbf{k}'}|}\Delta_{\mathbf{k}'}\int_{0}^{\xi_c} dx\frac{\tanh\left(|x|/2k_BT_c\right)}{|x|}.
\end{aligned}
\tag{107}
$$

Now we transform this integral back to a discrete sum:

$$
\Delta_{\mathbf{k}}=-\frac{1}{(2\pi)^d}\sum_{\mathbf{k}'}\frac{s_{\mathbf{k}'}}{|v_{\mathbf{k}'}|}V^{sing}_{\mathbf{k},\mathbf{k}'}\Delta_{\mathbf{k}'}\int_0^{\xi_c} dx\frac{\tanh\left(|x|/2k_BT_c\right)}{|x|}. \tag{108}
$$

Hence we define

$$
M_{\mathbf{k},\mathbf{k}'}=-\frac{1}{(2\pi)^d}\sum_{\mathbf{k}'}\frac{s_{\mathbf{k}'}}{|v_{\mathbf{k}'}|}V^{sing}_{\mathbf{k},\mathbf{k}'} \tag{109}
$$

and solve for the eigenvalue problem:

$$
\lambda\Delta_{\mathbf{k}'}=\sum_{k'}M_{\mathbf{k},\mathbf{k}'}\Delta_{\mathbf{k}'}. \tag{110}
$$

When

$$
\lambda=\int_0^{\xi_c} dx\frac{1}{|x|}\tanh\left(\frac{|x|}{2k_BT_c}\right) \tag{111}
$$

we have the superconductivity. Since ξ_c is small, we may approximate the integral to be

$$
\lambda=\frac{\xi_c\beta_c}{2}. \tag{112}
$$

Now I will justify why this equality is independent of ξ_c as long as it is small. First let us take a small ξ_c and acquire the matrix $M_{\mathbf{k},\mathbf{k}'}$ with eigenvalues $\lambda_1,...,\lambda_n$. Now we take $2\xi_c$. Since this change is very small, the new points for

$$
\Sigma=\{\mathbf{k}\in BZ:|\xi_{\mathbf{k}}|\leq 2\xi_c\}. \tag{113}
$$

are just a surface twice thicker than the original one. Since the thinkness is so small we may treat the new surface almost identical to the orginal one. It suffice to show with two points. Our new $\tilde{M}_{\mathbf{k},\mathbf{k}'}$ will be

$$
\begin{array}{|cc|cc|c|}
\hline
k_1 & k_2 & k_1' & k_2' & \\
\hline
A & B & A & B & k_1\\
C & D & C & D & k_2\\
\hline
A & B & A & B & k_1'\\
C & D & C & D & k_2'\\
\hline
\end{array}
=\begin{bmatrix} M & M\\ M & M\end{bmatrix}. \tag{114}
$$

Now I will prove the following statement

Theorem 1. *Let A have eigenvalues $\lambda_1,...,\lambda_n$ then*

$$
\begin{bmatrix} A & A\\ A & A\end{bmatrix} \tag{115}
$$

has eigenvalues $2\lambda_1,...,2\lambda_n,0,...,0$.

Proof. We prove by induction on the dimension of A. The base case if obvious. Assume the case holds for dimension n. Now let the dimension be $n+1$. First it is easy to verify that there are $n+1$ L.I. eigenvectors corresponding to eigenvalue 0: The $i-$th vector's $k-$th component is

$$
v_i^{(k)}=\delta_{k,i}-\delta_{k,i+(n+1)}. \tag{116}
$$

Then the trace of this large matrix is $2\mathrm{Tr}(A)=2\lambda_1+...+2\lambda_n+2\lambda_{n+1}$. This should equal $(2\lambda_1+0)+...+(2\lambda_n+0)+(\Lambda_{n+1}+0)$, so $\Lambda_{n+1}=2\lambda_{n+1}$.

□

Therefore the eigenvalues of $\tilde{M}_{\mathbf{k},\mathbf{k}'}$ will be doubled and $\Lambda/(2\xi_c) = \lambda/\xi_c$.

Meanwhile, we from the above theorem would wonder what would happen if we increase the $\mathbf{k}-$mesh. First of all, just like before we will have a larger matrix consisting of four identical blocks. However since

$$M_{\mathbf{k},\mathbf{k}'} = -\frac{1}{(2\pi)^d}\sum_{\mathbf{k}'}\frac{s_{\mathbf{k}'}}{|v_{\mathbf{k}'}|}V^{sing}_{\mathbf{k},\mathbf{k}'} \propto s_{\mathbf{k}'}, \tag{117}$$

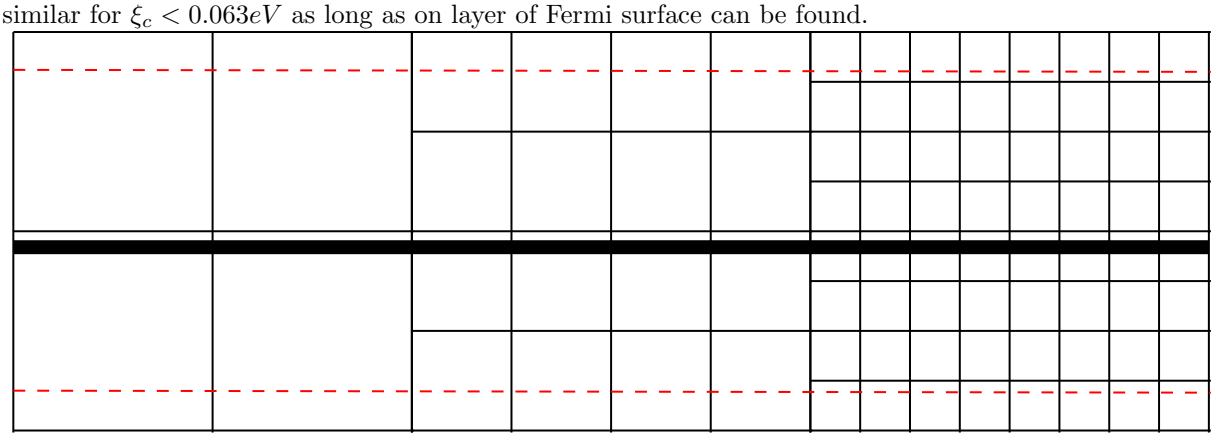
when we say, double the mesh, we will halve all the $s_{\mathbf{k}}$ and hence the matrix we get will be

$$\begin{bmatrix} M/2 & M/2 \\ M/2 & M/2 \end{bmatrix}. \tag{118}$$

So from the theorem we know that if the calculation has converged w.r.t. the $\mathbf{k}-$mesh, doubling the mesh again would only bring us another N_{FS} eigenvalues equal to 0. *However*, when numerically finding the Fermi surface, we have to set an energy threshold ξ_c and for fixed ξ_c, a greater $\mathbf{k}-$mesh could lead to a thicker Fermi surface sampling, quite equivalent to increasing ξ_c for fixed $\mathbf{k}-$mesh. This effect on the other hand, is limited. The maximum multiplier for our Fermi surface thickness is $1+2\xi_c/|v_F|$. So a secure way to acquire convergence is to increase indefinitely and divide each $s_{\mathbf{k}'}$ by $1+2\xi_c/|v_F|$.

Meanwhile, we must discuss how possible is such circumstance met. It turns out that this is hard. For our $d-s$ model the average $|v_F|$ is around $1eV\cdot a$. So a ξ_c at $0.02eV$ will correspond to a $0.02/a$ width in $\mathbf{k}-$space. Now a $\mathbf{k}-$mesh of 100 points/side corresponds to a grid of length $2\pi/100a \approx 0.063/a$. Therefore we might expect an increase in the thickness of FS when the $\mathbf{k}-$mesh is more than 300. But this is usually too high for computations.

Therefore although the eigenvalue of $M_{\mathbf{k},\mathbf{k}'}$ should be proportional to ξ_c, this relies on the Fermi surface's increased thickness. We just showed that an increase of the Fermi surface's thickness requires $0.063/a$ length increase of the threshold for a $\mathbf{k}-$mesh of 100. This corresponds to $\xi_c = 0.063eV$. Therefore the result should be similar for $\xi_c < 0.063eV$ as long as on layer of Fermi surface can be found.



Now we include the temperature dependence of $V(\mathbf{k},\mathbf{k}')$, which turns it into $V(k,k')$ by replacing $\chi_0(\mathbf{q})$ with $\chi_0(q) = \chi_0(i\Omega_m,\mathbf{q})$. The ultimate goal now for us is to solve

$$\lambda\Delta(k) = \sum_{k'} M(k,k')\Delta(k') \tag{119}$$

for $k \in FS$. The matrix M is given by

$$M(k,k') = -\frac{1}{2(2\pi)^d}\frac{s_{\mathbf{k}'}}{|\,v_{\mathbf{k}'|}}\left[V(k,k') \pm V(-k,k')\right] \tag{120}$$

For singlet state,

$$M(k,k') = -\frac{1}{2(2\pi)^d}\frac{s_{\mathbf{k}'}}{|v_{\mathbf{k}'}|}\left[2U + \frac{3}{2}U^2\chi_s(k-k') - \frac{1}{2}U^2\chi_c(k-k') + \frac{3}{2}U^2\chi_s(k+k') - \frac{1}{2}U^2\chi_c(k+k')\right] \tag{121}$$

To solve the equation numerically (discretely), we first calculate the $n = N_{FS}$ points on the fermi surface and label them by $\mathbf{k}_1, ..., \mathbf{k}_n$. Then the k points will be labeled by

$$(-2m\pi/\beta, \mathbf{k}_1), ..., (-2m\pi/\beta, \mathbf{k}_1), ..., (2m\pi/\beta, \mathbf{k}_n), ..., (2m\pi/\beta, \mathbf{k}_n). \tag{122}$$

Then the dimension of the matrix $M_{k,k'}$ will be $(2m+1)N_{FS}$.

4.3 Multi-orbital

Now let us consider the n-orbital case without temperature-dependent potential. In multiorbital case, a major difference is that we have two set of bases: One being the atomic basis, represented by $c_i^\dagger$, the other one being the eigenbasis $|i\rangle$. We denote the two set of bases by $\mathcal{A}$ and $\mathcal{E}$. The U-interaction is expressed under by U_{kl}^{ij}, which takes a direct form under $\mathcal{A}$. So $U_{kl}^{ij} = [U]_{\mathcal{A}\mathcal{A}}^{\mathcal{A}\mathcal{A}}$. However the original Hamiltonian is labeled by index of the eigenbasis, so the Γ matrix we will be using must be expressed under $\mathcal{E}$. So $\Gamma_{kl}^{ij} = [\Gamma]_{\mathcal{E}\mathcal{E}}^{\mathcal{E}\mathcal{E}}$. Since the Γ matrix (representing the potential) involves U, we must do basis transform to do multiplications and additions:

$$[\Gamma]_{\mathcal{E}\mathcal{E}}^{\mathcal{E}\mathcal{E}} = [I]_{\mathcal{E}\mathcal{E}}^{\mathcal{E}\mathcal{A}}[I]_{\mathcal{E}\mathcal{E}}^{\mathcal{A}\mathcal{E}}[V]_{\mathcal{A}\mathcal{A}}^{\mathcal{A}\mathcal{A}}[I]_{\mathcal{E}\mathcal{A}}^{\mathcal{E}\mathcal{E}}[I]_{\mathcal{A}\mathcal{E}}^{\mathcal{E}\mathcal{E}}. \tag{123}$$

First we have a potential V under $\mathcal{A}$ representation given by

$$\tilde{V}_{l_1l_2,l_3l_4} = \tilde{V}_{l_4l_3}^{l_2l_1}(\mathbf{q}) = \left[U + \frac{3}{2}U\chi_s(\mathbf{q})U - \frac{1}{2}U\chi_c(\mathbf{q})U\right]_{l_4l_3}^{l_2l_1}. \tag{124}$$

this is defined on the entire BZ.

$$V_{ij,kl} = V_{lk}^{ji}(\mathbf{k},\mathbf{k}') = \sum_{l_1l_2l_3l_4} a_k^{l_3,*}(-\mathbf{k})a_l^{l_4,*}(\mathbf{k})\,\mathrm{Re}\left[\tilde{V}_{l_4l_3}^{l_2l_1}(\mathbf{k},\mathbf{k}')\right]a_i^{l_1}(\mathbf{k}')\,a_j^{l_2}(-\mathbf{k}'). \tag{125}$$

Our $\Gamma_{ij} = V_{jj,ii} = \Gamma_{ii}^{jj}$ so

$$\Gamma_{ij}(\mathbf{k},\mathbf{k}') = \sum_{l_1l_2l_3l_4} a_i^{l_3,*}(-\mathbf{k})a_i^{l_4,*}(\mathbf{k})\,\mathrm{Re}\left[\tilde{V}_{l_4l_3}^{l_2l_1}(\mathbf{k},\mathbf{k}')\right]a_j^{l_1}(\mathbf{k}')\,a_j^{l_2}(-\mathbf{k}'). \tag{126}$$

Since $\mathbf{k} \in FS_i$ and $\mathbf{k}' \in FS_j$. This tensor for a two-orbital model can be arranged in the form of

$$\begin{bmatrix}\Gamma_{11} & \Gamma_{12}\\ \Gamma_{21} & \Gamma_{22}\end{bmatrix}, \tag{127}$$

where Γ_{11} is a $|FS_1| \times |FS_1|$ matrix with index labeling the $(\mathbf{k},\mathbf{k}')$ points, and Γ_{12} a $|FS_1| \times |FS_2|$ matrix... Hence the size of this Γ tensor should be $|FS|^2$.

From our BCS equation

$$\Delta_{\mathbf{k},i} = -\sum_{\mathbf{k}',j}\Gamma_{ij}\frac{\Delta_{\mathbf{k}',j}}{2E_{\mathbf{k}',j}}\tanh\left(\frac{E_{\mathbf{k}',j}}{2k_BT}\right) \tag{128}$$

we can again apply the Eliashberg linearization:

$$\Delta_{\mathbf{k},i} = -\frac{1}{(2\pi)^d}\sum_{\mathbf{k}',j}\frac{s_{\mathbf{k}',j}}{|v_{\mathbf{k}',j}|}\Gamma_{ij}(\mathbf{k},\mathbf{k}')\Delta_{\mathbf{k}',j}\int_0^{\xi_c}dx\frac{\tanh(|x|/2k_BT_c)}{|x|}. \tag{129}$$

So now we may define the $M_{\mathbf{k},\mathbf{k}'}$ matrix to be

$$M_{\mathbf{k},\mathbf{k}'} := -\frac{1}{(2\pi)^d}\frac{s_{\mathbf{k}',j}}{|v_{\mathbf{k}',j}|}\Gamma_{ij}(\mathbf{k},\mathbf{k}'). \tag{130}$$

Then by solving $M_{\mathbf{k},\mathbf{k}'}\Delta_{\mathbf{k}'} = \lambda\Delta_{\mathbf{k}}$, we identify

$$\lambda = \int_0^{\xi_c}dx\frac{\tanh(|x|/2k_BT_c)}{|x|} \tag{131}$$

again.

The U matrix corresponds to the interaction term

$$U_{l_1 l_2 l_3 l_4} c^\dagger_{l_1,\sigma} c_{l_2,\sigma} c^\dagger_{l_3,\sigma'} c_{l_4,\sigma'}. \tag{132}$$

Now let us look at an example of a two-orbital model consisting of $d_{x^2-y^2}$ and s orbitals on two different sites. We neglect the Columb interation on the s orbital. The interorbital repulsion on the $d_{x^2-y^2}$ is U. So the interaction term is

$$U_{\text{int}} = U n_{1\uparrow} n_{1\downarrow}. \tag{133}$$

Therefore the U matrix in this system is

$$U_{\text{Ni},l_1 l_2 l_3 l_4} = \begin{cases} U, & \text{if } l_1 = l_2 = l_3 = l_4 = 1 \\ 0, & otherwise. \end{cases} \tag{134}$$

For another example of a two-orbital model consisting of $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals on one site. Now the U matrix becomes

$$U_{\alpha,l_1 l_2 l_3 l_4} = \begin{cases} U_\alpha & l_1 = l_2 = l_3 = l_4, \\ U'_\alpha & l_1 = l_3 \neq l_2 = l_4, \\ J_\alpha & l_1 = l_2 \neq l_3 = l_4, \\ J'_\alpha & l_1 = l_4 \neq l_2 = l_3. \end{cases} \tag{135}$$

More specifically,

l_1	l_2	l_3	l_4	$U_{l_1 l_2 l_3 l_4}$
1	1	1	1	U
2	2	2	2	U
1	2	1	2	$U-2J$
2	1	2	1	$U-2J$
1	1	2	2	J
2	2	1	1	J
1	2	2	1	J
2	1	1	2	J

(136)

Now we investigate a special type of Hamiltonian

$$H(\mathbf{k}) = \begin{bmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{bmatrix}. \tag{137}$$

It is already diagonal so $a = \delta$. Then χ_0 should be

$$\chi_0(\mathbf{q})^{i_2 i_1}_{i_4 i_3} = -\frac{1}{N_\mathbf{k}} \sum_\mathbf{k} \sum_{j_1,j_2} \delta^{i_1}{}_{j_1} \delta_{i_3}{}^{j_1} \delta_{i_4}{}^{j_2} \delta^{i_2}{}_{j_2} \frac{n_F[(\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1}] - n_F[(\epsilon_\mathbf{k})_{j_2}]}{(\epsilon_{\mathbf{k}+\mathbf{q}})_{j_1} - (\epsilon_\mathbf{k})_{j_2}} \tag{138}$$

$$= -\frac{1}{N_\mathbf{k}} \sum_\mathbf{k} \frac{n(\mathbf{k}+\mathbf{q})_{i_1} - n(\mathbf{k})_{i_2}}{(\epsilon_{\mathbf{k}+\mathbf{q}})_{i_1} - (\epsilon_\mathbf{k})_{i_2}} = \chi_0(\mathbf{q})^{i_2 i_1}_{i_2 i_1}. \tag{139}$$

What we are left with are just $\chi_0(\mathbf{q})^{11}_{11}, \chi_0(\mathbf{q})^{22}_{22}, \chi_0(\mathbf{q})^{12}_{12}$ and $\chi_0(\mathbf{q})^{21}_{21}$.

Now, back to

$$\Gamma_{ij}(\mathbf{k},\mathbf{k}') = \sum_{l_1 l_2 l_3 l_4} \delta_i^{l_3} \delta_i^{l_4} \operatorname{Re}\left[\tilde{V}^{l_2 l_1}_{l_4 l_3}(\mathbf{k},\mathbf{k}')\right] \delta_j^{l_1} \delta_j^{l_2} = \tilde{V}^{jj}_{ii}(\mathbf{k},\mathbf{k}'). \tag{140}$$

Now back to

$$\tilde{V}^{jj}_{ii}(\mathbf{q}) = U^{jj}_{ii} + \frac{3}{2}[U\chi_s(\mathbf{q})U]^{jj}_{ii} - \frac{1}{2}[U\chi_c(\mathbf{q})U]^{jj}_{ii}. \tag{141}$$

s